

RAM KRISHNA YADAV

FULL STACK SOFTWARE DEVELOPER

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SUMMARY

Full Stack Software Engineer with extensive experience architecting high-performance, production-grade web systems. Creator of **Zyotra**, an **agentless automated deployment PaaS**, and author of an open-source **JavaScript thread-pool library** with **10,000+ npm downloads**. Deep expertise in designing scalable REST APIs, **role based systems**, **WebSocket real-time architectures**, and concurrency models using TypeScript, Node.js, and PostgreSQL. Proven ability to build complex distributed systems from concept to production.

SKILLS

- Languages : Javascript, Typescript, Python, C++
- Frontend : React, Redux Toolkit, Tailwind CSS, HTML5
- Backend : Node.js, Express.js, Elysia.js, REST APIs, JWT, WebSockets
- Databases : PostgreSQL, MongoDB, MySQL, ORMs
- Systems and Tools : Worker Threads, Event-Driven Architecture, **Docker**, **AWS**, **VPS Deployment**, Git, Jest Testing

PROFESSIONAL EXPERIENCE

MERN Stack Intern at Blue Fox Pvt. Ltd. | Oct 2024-Dec 2024

- Contributed to backend and frontend development of **SaaS applications** across hotel, e-commerce, and restaurant domains.
- Implemented RESTful APIs with JWT-based authentication and **role-based access** control.

PROJECTS

Zyotra — Self-Hosted PaaS & Automated Deployment Platform [Live](#) | [Github](#)

Tech : TypeScript, Bun, Node.js, Nginx, PostgreSQL, MySQL, Redis, WebSockets, AWS S3, Docker

- Built a self-hosted PaaS replicating the Vercel/Heroku experience on bare-metal servers, cutting cloud infrastructure costs by up to **80%**
- Engineered **agentless deployment automation** via SSH scripts handling Git cloning, SSL termination (Certbot), and **Nginx reverse proxy configuration**
- Architected **real-time build log streaming** and **machine analytics** (CPU/RAM) via WebSockets
- Built a **Database Management System** with PostgreSQL/MySQL backups to S3, Redis deployment, and an integrated **SQL Query Runner**

Thready.js — Open-source JavaScript Thread Pool Library [Live](#) | [Github](#)

Tech : TypeScript, NPM Package

- Designed a type-safe **thread-pool abstraction** for Web Workers and Node.js worker_threads
- Implemented **worker pooling**, **task queueing**, and a **zero-dependency** architecture
- Abstracted low-level concurrency patterns into a developer-friendly API, demonstrating strong understanding of JavaScript's **event loop and worker model**
- Achieved **10,000+ npm downloads** , indicating real-world adoption

ChessOnline - Realtime Multiplayer Chess Game [Live](#) | [Github](#)

Tech : React.js, Node.js, Express.js, WebSocket, MongoDB, Tailwind CSS, TypeScript

- Built a real-time multiplayer chess application with **low-latency WebSocket communication**
- Implemented **private rooms**, turn-based gameplay, and game-state consistency
- Developed a responsive chessboard UI validation, and complete rule enforcement
- Designed a scalable backend to **manage concurrent game sessions** and **real-time player matching**.

CRM System - Medical Equipment Distribution Platform [Live](#) | [Github](#)

Tech : Next.js , TypeScript, Express.js, PostgreSQL, Prisma ORM, Redux Toolkit, JWT

- Developed a **multi-role CRM system** with JWT authentication and **role-based access** for admins, distributors, and customers
- Implemented full order lifecycle management with PDF invoice generation
- Designed scalable REST APIs using PostgreSQL and Prisma with **advanced search and filtering**
- Built an admin **analytics dashboard** for sales, inventory, and order insights

XCodeGen-AI Assistant for Efficient Coding [Live](#) | [Github](#)

Tech : React.js, Node.js, Express.js, MongoDB, Typescript, Llama, Gemini, Mistral

- Built an AI-powered developer assistant for **automated code generation** and documentation
- Implemented **Developer Mode and Chat Mode** to support project scaffolding, refactoring, and contextual assistance
- Integrated a controlled **in-app file system** enabling safe file-level modifications by the LLM
- Leveraged multiple LLMs (Llama, Gemini, Mistral, Qwen) to improve response quality across diverse tasks

EDUCATION

Bachelor of Computer Engineering (2024–2028)

Tribhuvan University, IOE Purwanchal Campus, Nepal